

# MATH 174 - Numerical Analysis

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# 1 Error

## 1.1 Approximation

Let  $x$  be an exact number.

Let  $y$  be an approximation of  $x$ .

**Def:** The absolute error is  $|x - y|$ .

**Def:** The relative error is  $\frac{|x - y|}{|x|}$ ,

## 1.2 Round-off Error

Usually a machine uses a finite amount of memory to store a number.

machine  $\leftarrow \pi$

Suppose a machine can store some number of digits.

1. Allocate a digit for sign, positive or negative.
2. Allocate a number of digits for the mantissa.
3. Allocate a digit for sign of the exponent.
4. Allocate a number of digits for the exponent.

Under rounding in the case of a 4 digit mantissa,  $\pi = 0.3142 \cdot 10^1$ , this is the machine # representation of  $\pi$ .

**Def:** Round-off error are all errors arising from the error between actual # and machine #.

**Notation:**

$$x = \text{real \#}$$

$$fl(x) = \text{machine \# representation of } x$$

In an s-digit rounding machine,

$$\text{Absolute error} = |x - fl(x)|$$

$$\text{Relative error} = \frac{|x - fl(x)|}{|x|} \leq \frac{1}{2} \cdot 10^{1-5} = 5 \cdot 10^{-5}$$

### 1.3 Floating Point Arithmetic

What happens under arithmetic operations? (+, -, ·, /)

Two things to watch out for:

1. Many operations can degrade error.
2. Catastrophic cancellation.

#### 1.3.1 Degrading Error

|         |                     |
|---------|---------------------|
| number  | machine             |
| $x$     | $fl(x)$             |
| $y$     | $fl(y)$             |
| $x + y$ | $fl(fl(x) + fl(y))$ |

The more nested  $fl()$  there are, the larger the error will be. The same follows for -, ·, /.

When performing arithmetic, it temporarily allows for more digits than the mantissa allows.

Algorithmic "solutions"

$$((x + y) + z) + w$$

Worst case, 4 times the error.

$$(x + y) + (z + w)$$

Worst case, 3 times the error.

### 1.3.2 Catastrophic Cancellation

**Ex:** 3-digit rounding machine. Given 2 numbers: 0.312 and 0.311.

$$0.312 - 0.311 = 0.001 = 0.1 \cdot 10^{-2}$$

Loss of significant digits!

**Ex:** 5-digit rounding machine. Given 2 numbers:

$$x = 0.41626324$$

$$y = 0.41626323$$

$$x - y = 0.00000001$$

In machine:

$$fl(x) = 0.41626$$

$$fl(y) = 0.41626$$

$$fl(x) - fl(y) = 0.00000 = 0$$

$$fl(fl(x) - fl(y)) = 0$$

Relative error:

$$\begin{aligned} \frac{|x - y - fl(fl(x) - fl(y))|}{|x - y|} &= \frac{|x - y - 0|}{|x - y|} \\ &= 1 = 100\% \text{ error!} \end{aligned}$$

## 2 Speed

### 2.1 Dot Product

Consider dot product of vectors

$$\vec{x} \in \mathbb{R}^n, \vec{y} \in \mathbb{R}^n$$

$$\begin{aligned} \vec{x} \cdot \vec{y} &= \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \\ &= x_1 y_1 + x_2 y_2 + \dots + x_n y_n \end{aligned}$$

Count time-consuming operations:

1. Add/subtract
2. Multiply/divide
3. Comparisons
4. Memory access/storage

These are called FLOPS (floating point operations).

For the dot product, there are  $n$  multiplications and  $n-1$  additions.

Adding these together, we get  $2n - 1$  FLOPS.

*Note that this expression is linear in  $n$ .*

For large  $n$ ,  $2n$  dominates,  $= \mathcal{O}(n) \approx \text{constant times } n$ .

Ex: For  $n$  vs  $2n$ ,  $2n$  will take twice as long.

Ex: For  $\mathcal{O}(n^2)$ ,  $n$  vs  $2n$ ,  $2n$  will take four times as long.

## 2.2 Memory

Consider a matrix  $A_{n \times n}$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \text{ vs } \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

The square matrix uses  $n^2$  memory locations, while the vector uses  $n$  memory locations. Each number also uses 8 bytes, creating another goal for algorithm development, reduce memory usage.

## 3 Gaussian Elimination

### 3.1 Triangular Matrices

Write a MATLAB function that when given  $A\vec{x} = \vec{b}$ , computes  $\vec{x}$

$$A\vec{x} = \vec{b}$$

A is triangular and invertible

Let's say A is lower triangular.  $a_{ij} = 0, \forall j > i$

$$\begin{bmatrix} a_{11} & 0 & 0 & \dots & 0 \\ a_{21} & a_{22} & 0 & \dots & 0 \\ a_{31} & a_{32} & a_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix}$$

Similarly, if A is upper triangular,  $a_{ij} = 0, \forall j < i$ .

**Ex:**  $A\vec{x} = \vec{b}$ , A is upper triangular.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

This gives us the following system of equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{33}x_3 &= b_3 \end{aligned}$$

So  $x_3 = \frac{b_3}{a_{33}}, a_{33} \neq 0$  if A is invertible.

Then,  $x_2 = \frac{b_2 - a_{23}x_3}{a_{22}}, a_{22} \neq 0$  if A is invertible.

Finally,  $x_1 = \frac{b_1 - a_{12}x_2 - a_{13}x_3}{a_{11}}, a_{11} \neq 0$  if A is invertible.

This is called back substitution.

For  $i = n, n-1, \dots, 1$

$$\begin{aligned} x_i &= \frac{b_i - (a_{i,i+1}x_{i+1} + \dots + a_{i,n}x_n)}{a_{ii}} \\ &= \frac{b_i - \sum_{j=i+1}^n a_{ij}x_j}{a_{ii}} \end{aligned}$$

**Ex:**  $A\vec{x} = \vec{b}$ , A is lower triangular.

$$\begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

This gives us the following system of equations:

$$\begin{aligned}a_{11}x_1 &= b_1 \\a_{21}x_1 + a_{22}x_2 &= b_2 \\a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3\end{aligned}$$

This is called forward substitution.

For  $i = 1, 2, \dots, n$

$$\begin{aligned}x_i &= \frac{b_i - (a_{i,1}x_1 + \dots + a_{i,i-1}x_{i-1})}{a_{ii}} \\&= \frac{b_i - \sum_{j=1}^{i-1} a_{ij}x_j}{a_{ii}}\end{aligned}$$

Now given a regular matrix, make it triangular without changing the solutions. Recall Gaussian elimination.

1.  $E_i$  (equation  $i$ ) replaced by  $cE_i$ . Row scaling.
2.  $E_i$  swapped with  $E_j$ . Row swapping.
3.  $E_i$  replaced by  $E_i + cE_j$ . Row addition.

### 3.1.1 Speed of Back Substitution

For  $i = n, n-1, \dots, 1$

$$x_i = \frac{b_i - \sum_{j=i+1}^n u_{ij}x_j}{u_{ii}}$$

|          |              |
|----------|--------------|
| $i = n$  | 1 flop       |
| $n-1$    | 3 flops      |
| $\vdots$ | $\vdots$     |
| $1$      | $2n-1$ flops |

$$\sum_{j=1}^n (2j-1) = 2 \sum_{j=1}^n (j) - \sum_{j=1}^n 1 = 2 \left( \frac{n(n+1)}{2} \right) - n = n^2$$

With a large  $n$ , the term  $n^2$  dominates, meaning  $\mathcal{O}(n^2)$

## 3.2 Augmented Matrices

Gaussian Elimination on same matrices and different RHS comes up often in applications and algorithms.

### 3.2.1 LU Factorization

$$A = LU$$

where L is lower triangular with 1's in the diagonal and U is upper triangular

The important entries of L are called multipliers. These entries can be found with the negative of the coefficients of the steps of Gaussian elimination.

$$\begin{array}{ll} L\vec{y} = \vec{b} & \text{get y from forward substitution} \\ U\vec{x} = \vec{y} & \text{get x from back substitution} \end{array}$$

## 3.3 Matrix Inverses

Matrix-vector multiplication:

$$(A\vec{x})_i = \sum_{j=1}^n a_{ij}x_j$$

which gives  $2(n-1)n$  flops total

Backwards substitution uses  $n^2$  flops.

Forward substitution uses  $n^2 - n$  flops. Because we're working with L, which has 1's on the diagonal, we save  $n$  divisions.

In summary, getting  $A^{-1}$  to solve  $A\vec{x} = \vec{b}$  is slower. Also,  $A^{-1}$  has memory storage issues for certain matrices. For example, a tridiagonal matrix, even though it has zeroes on most of its entries,  $A^{-1}$  has no zeroes.



### 3.3.1 Gaussian Elimination FLOP count

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix}$$

This step has 1 division, 1 multiplication, 1 addition/subtraction.

In total there are  $2n - 1$  total flops.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ 0 & a_{22} & a_{23} & \dots & a_{2n} \\ 0 & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix}$$

Repeat this for  $n - 1$  rows (minus 1 because we are excluding the first row).

We can then repeat this for the inner block matrix.

$$(2n - 1)(n - 1) + (2(n - 1) - 1)((n - 1) - 1) + (2(n - 2) - 1)((n - 2) - 1) \dots$$

$$\sum_{j=1}^n j = \frac{n(n + 1)}{2}$$

$$\sum_{j=1}^n j^2 = \frac{n(n + 1)(2n + 1)}{6}$$

## 4 Gaussian Elimination With Pivoting

But what about  $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  Gaussian elimination fails on first step. Can we use another elementary row operation, namely, row swapping?

In general,

$$\begin{bmatrix} 0 & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix}$$

## 4.1 Error

Find a nonzero element  $a_{i1}$  then perform  $E_1 \leftrightarrow E_i$ . We have to do this before each elimination step. However, there is error involved.

$$\begin{aligned} \left[ \begin{array}{cc|c} \epsilon & 1 & 1 \\ \times & \times & \times \end{array} \right] &\rightarrow \tilde{x}_2 = x_2 + \delta \\ &\rightarrow \tilde{x}_1 = \frac{1 - \tilde{x}_2}{\epsilon} = \frac{1 - x_2 - \delta}{\epsilon} = \frac{1 - x_2}{\epsilon} - \frac{\delta}{\epsilon} \end{aligned}$$

Note that in this situation, choosing an  $x_1$  that is small leads to a small  $\epsilon$  which leads to a large error. We have to be careful when swapping rows.

$$\left[ \begin{array}{cc|c} \epsilon & 1 & 1 \\ 0 & \times & \times \end{array} \right]$$

$$\begin{aligned} \tilde{x}_2 &= x_2 + \delta \\ |\delta| &= |\tilde{x}_2 - x_2| \\ \tilde{x}_1 &= x_1 - \frac{\delta}{\epsilon} \\ \left| \frac{\delta}{\epsilon} \right| &= |\tilde{x}_1 - x_1| \end{aligned}$$

#### 4.1.1 Section 7.5

$$\begin{aligned}A\vec{x} &= \vec{b} \\A(\vec{x} + \vec{\delta x}) &= \vec{b} + \vec{\delta b} \\A\vec{\delta x} &= \vec{\delta b} \\\vec{\delta x} &= A^{-1}\vec{\delta b}\end{aligned}$$

We can use a norm to measure the size of error vectors.

$$\begin{aligned}\|\vec{\delta x}\| &= \|A^{-1}\vec{\delta b}\| \leq \|A^{-1}\| \|\vec{\delta b}\| \\\frac{\|\vec{\delta x}\|}{\|\vec{x}\|} &\leq \|A\| \|A^{-1}\| \frac{\|\vec{\delta b}\|}{\|\vec{b}\|} \\\text{relative error} &\leq \text{condition \#} \times \text{relative size of change of } \vec{b}\end{aligned}$$

**Ex:** Suppose you need to solve accurately  $A\vec{x} = \vec{b}$  and someone gives you the approx  $\tilde{x}$ , is it accurate?

We can use the residual vector.

$$\begin{aligned}\vec{r} &= \vec{b} - A\tilde{x} \\A\tilde{x} &= \vec{b} - \vec{r}\end{aligned}$$

If A is well-conditioned (small condition number), then  $\frac{\|\vec{r}\|}{\|\vec{b}\|}$  is small and  $\frac{\|\vec{x} - \tilde{x}\|}{\|\vec{x}\|}$  small.

If A is ill-conditioned (large condition number), then we don't know.

## 4.2 Row Swapping

$$PA = LU$$

where PA is A with rows swapped. P is the identity matrix with its rows swapped to denote the row swapping operations done on A.

Now solving LU factorization:

$$\begin{aligned}A\vec{x} &= \vec{b} \\PA\vec{x} &= P\vec{b} \\LU\vec{x} &= P\vec{b} \\L\vec{y} &= P\vec{b} \quad \text{forward sub} \\U\vec{x} &= \vec{y} \quad \text{back sub}\end{aligned}$$

### 4.3 Speed

In an  $n \times n$  matrix, it takes  $n - 1$  comparisons for the first step,  $n - 2$  for the second step, and so on.

$$\begin{aligned}\sum_{j=1}^{n-1} j &= \frac{(n-1)n}{2} \\&= \mathcal{O}(n^2)\end{aligned}$$

This is negligible additional amount of work for large  $n$  compared to  $\mathcal{O}(n^3)$  of Gaussian elimination.

Even safer than row partial pivoting in terms of error:

1. Scaled row partial pivoting:
2. Complete pivoting: row swaps and column swaps to move the largest element to pivot element.

## 5 Approximations

### 5.1 Fixed Point Problems

Solve an equation  $g(x) = h(x)$

Modify to  $f(x) = x$

Now we have the fixed point problem, which we can try to solve using fixed point iterations.

Start with guess  $x^{(0)}$

Then iterate  $x^{(k+1)} = f(x^{(k)})$

The  $x^{(k)}$  are approximations of the fixed point.

We can also write  $A\vec{x} = \vec{b}$  as fixed point problem using splitting.

$$A = M - N$$

$$(M - N)\vec{x} = \vec{b}$$

$$M\vec{x} = N\vec{x} + \vec{b}$$

$$\vec{x} = M^{-1}(N\vec{x} + \vec{b})$$

where the RHS is  $f(\vec{x})$

Choices for M, N:

Let  $A = D - L - U$ , where D is diagonal, L is lower triangular, U is upper triangular. We have

1. Jacobi iterative method.
2. Gauss-Seidel iterative method.

### 5.1.1 Jacobi Iterative Method

$$M = D$$

$$N = L + U$$

$$D\vec{x} = (L + U)\vec{x} + \vec{b}$$

$$x_i^{(k+1)} = \frac{b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k)} - \sum_{j=i+1}^n a_{ij}x_j^{(k)}}{a_{ii}}$$

### 5.1.2 Gauss-Seidel Iterative Method

$$M = D - L$$

$$N = U$$

$$(D - L)\vec{x} = U\vec{x} + \vec{b}$$

$$x_i^{(k+1)} = \frac{b_i - \sum_{j>i} a_{ij}x_j^{(k)} - \sum_{j<i} a_{ij}x_j^{(k+1)}}{a_{ii}}$$

## 5.2 Convergence

Jacobi & Gauss-Seidel are iterative methods. They start with a guess, which improves their approximations at each step. We want  $\vec{x}^{(k)}$  sequence of approximations to converge to the exact solution.

Practically, we need a stopping condition. Here are our choices:

1. Stop after certain number of steps
2. Stop when residual is small, where the residual is  $\vec{r} = \vec{b} - A\vec{x}$ ,  $\|\vec{r}\| \leq \text{tolerance}$ .
3. Stop when the algorithm slows down,  $\|x^{(k+1)} - x^{(k)}\| \leq \text{tolerance}$ .

$$A = M - N$$

$$\text{Jacobi} \quad M = D$$

$$\text{Gauss-Seidel} \quad M = D - L$$

$$\begin{aligned} A\vec{x} = \vec{b} &\rightarrow (M - N)\vec{x} = \vec{b} \\ &\rightarrow M\vec{x} = N\vec{x} + \vec{b} \\ &\rightarrow \vec{x} = M^{-1}(N\vec{x} + \vec{b}) \\ &\rightarrow \vec{x}^{(k+1)} = M^{-1}(N\vec{x}^{(k)} + \vec{b}) \\ &\rightarrow \vec{x}^{(k+1)} = \mathbf{T}\vec{x}^{(k)} + \vec{c} \end{aligned}$$

where  $\mathbf{T}$  is the iteration matrix,  $M^{-1}N$

and  $\vec{c}$  is  $M^{-1}\vec{b}$

An iterative method using  $\vec{x}^{(k+1)} = \mathbf{T}\vec{x}^{(k)} + \vec{c}$  converges if  $\varphi(\mathbf{T}) < 1$ , where  $\varphi(\mathbf{T})$  is the spectral radius of  $\mathbf{T}$ .

**Def:** Spectral radius  $\varphi(A) = \max\{|\lambda_1|, |\lambda_2|, \dots, |\lambda_n|\}$

$$\begin{aligned} \|\vec{e}^{k+1}\| &\approx \varphi(T) \|\vec{e}^k\| \\ \|\vec{e}^k\| &\approx \varphi(T)^k \|\vec{e}^{(0)}\| \end{aligned}$$

Let  $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$

$$T_J = \begin{bmatrix} 0 & \frac{-b}{a} \\ \frac{-b}{c} & 0 \end{bmatrix}$$

$$T_{GS} = \begin{bmatrix} 0 & \frac{-b}{a} \\ 0 & \frac{b^2}{ac} \end{bmatrix}$$

$$\det T_J = \begin{bmatrix} -\lambda & \frac{-b}{a} \\ \frac{-b}{c} & -\lambda \end{bmatrix} = \lambda^2 - \frac{b^2}{ac}$$

$$\det T_{GS} = \begin{bmatrix} -\lambda & \frac{-b}{a} \\ 0 & \frac{b^2}{ac} - \lambda \end{bmatrix} = \lambda(\lambda - \frac{b^2}{ac})$$

$$\lambda_J = \pm \sqrt{\frac{b^2}{ac}} = \varphi(T_J)$$

$$\lambda_{GS} = 0 \text{ or } \frac{b^2}{ac}, \quad \varphi(T_{GS}) = \frac{b^2}{ac}$$

If they converge, then  $\frac{b^2}{|ac|} < 1$ , and when this condition is true, Gauss-Seidel's method has a lower spectral radius than Jacobi, since  $x < \sqrt{x}$  for  $x < 1$ .

However, there are examples for general matrices where Jacobi's method converges and Gauss-Seidel's method does not. So for general matrices, we do not know which one is faster. In practice, however, Gauss-Seidel is almost always faster.

Other iterative methods:

1. Richardson's
2. Successive Over-Relaxation (SOR) - Gauss-Seidel is in this class
3. Conjugate Gradient (not splitting)
4. GMRES

### 5.2.1 Richardsons' Iterative Method

$$\begin{aligned}
A\vec{x} &= \vec{b} \\
\omega A\vec{x} &= \omega\vec{b} \\
\omega A &= I - I + \omega A \\
&= I - (I - \omega A) \text{ where } I \text{ is } M, \text{ and } (I - \omega A) \text{ is } N
\end{aligned}$$

$$\begin{aligned}
x^{k+1} &= M^{-1}(Nx^k + \omega b) \\
&= Nx^k + \omega b \\
&= (I - \omega A)x^k + \omega b \\
x_i^{k+1} &= x_i^k - \omega \sum_{j=1}^n a_{ij}x_j^k + \omega b_i
\end{aligned}$$

## 6 Eigenvalues

### 6.1 Finding Eigenvalues of a Matrix

Given  $A \in \mathbb{R}^{n \times n}$ , we want to find either an eigenvalue or all eigenvalues.

Suppose we have  $\lambda_1, \lambda_2, \dots, \lambda_n$ , where  $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$

Consider the special case where  $A$  has a linearly independent set of eigenvectors (which is true when  $A$  is symmetric).

Let our eigenvectors be  $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$  with  $\vec{x}_i$  corresponding to  $\lambda_i$ .

So  $A\vec{x}_i = \lambda_i\vec{x}_i$ , with the iterative method, we can start with initial guess  $\vec{x}^{(0)}$

$$\begin{aligned}
\vec{x}^{(0)} &= \alpha_1\vec{x}_1 + \alpha_2\vec{x}_2 + \dots + \alpha_n\vec{x}_n \\
A\vec{x}^{(0)} &= A\alpha_1\vec{x}_1 + A\alpha_2\vec{x}_2 + \dots + A\alpha_n\vec{x}_n \\
A\vec{x}^{(0)} &= \alpha_1A\vec{x}_1 + \alpha_2A\vec{x}_2 + \dots + \alpha_nA\vec{x}_n \\
A\vec{x}^{(0)} &= \alpha_1\lambda_1\vec{x}_1 + \alpha_2\lambda_2\vec{x}_2 + \dots + \alpha_n\lambda_n\vec{x}_n \\
A^k\vec{x}^{(0)} &= \alpha_1\lambda_1^k\vec{x}_1 + \alpha_2\lambda_2^k\vec{x}_2 + \dots + \alpha_n\lambda_n^k\vec{x}_n \\
&\approx \alpha_1\lambda_1^k\vec{x}_1
\end{aligned}$$